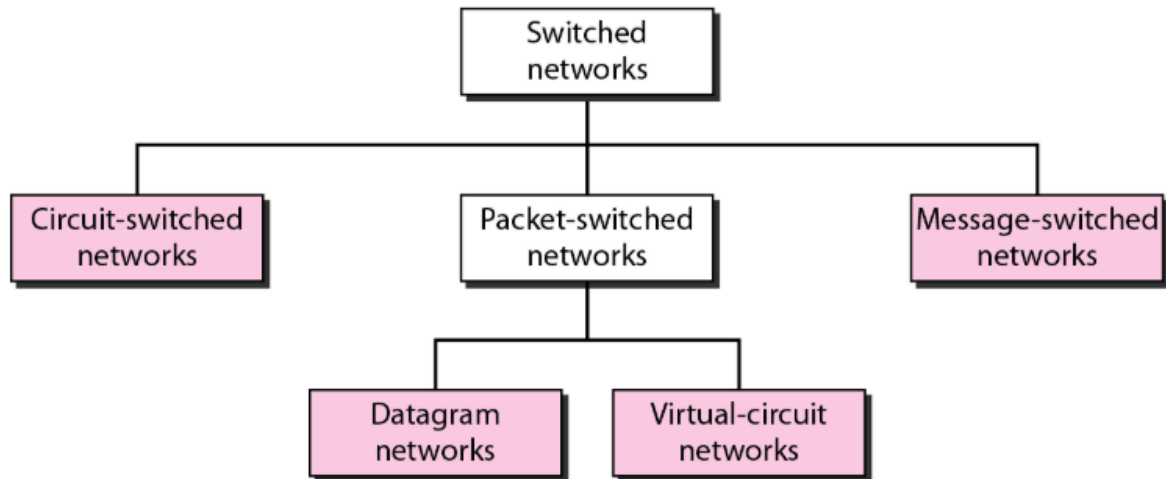


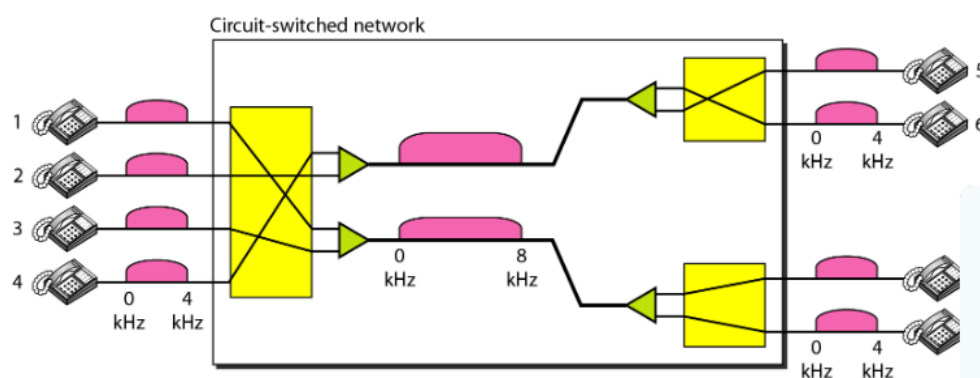
Day 13 Of Communication System Class

2. Switching



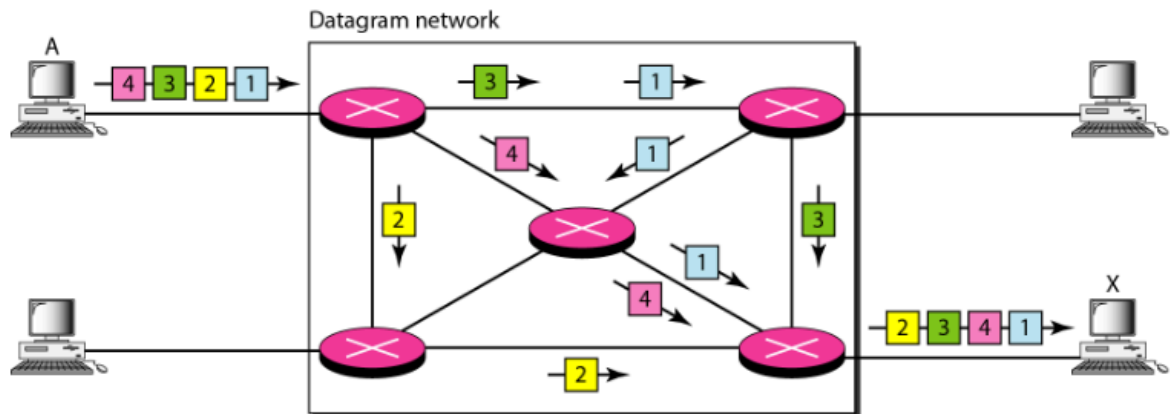
2.1 Circuit Switching

In circuit switching, the resources need to be reserved during the setup phase; the resources remain dedicated for the entire duration of data transfer until the teardown phase.



2.2 Packet Switching

In a packet-switched network, there is no resource reservation; resources are allocated on demand.



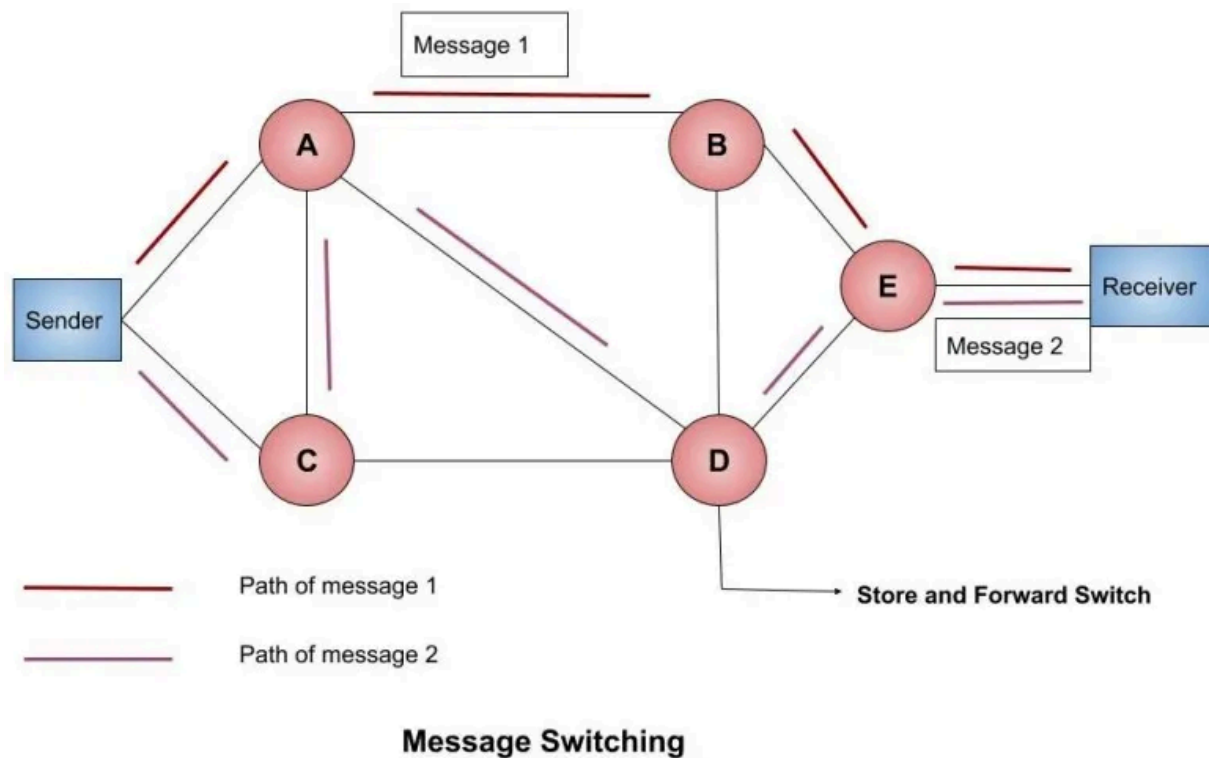
Switching in the Internet is done by using the datagram approach to packet switching at the network layer.

1. Compare packet switching and message switching.

[Most Probable Question]

Circuit Switching	Packet Switching
Physical path between source and destination	No physical path
All packets use same path	Packets travel independently
Reserve the entire bandwidth in advance	Does not reserve
Bandwidth Wastage	No Bandwidth wastage
No store and forward transmission	Supports store and forward transmission

2.3 Message Switching



3. Multiple Access

Channelization is a multiple-access method in which the available bandwidth of a link is shared in time, frequency, or through code, between different stations. In this section, we discuss three channelization protocols.

Topics discussed in this section:

Frequency-Division Multiple Access (FDMA)

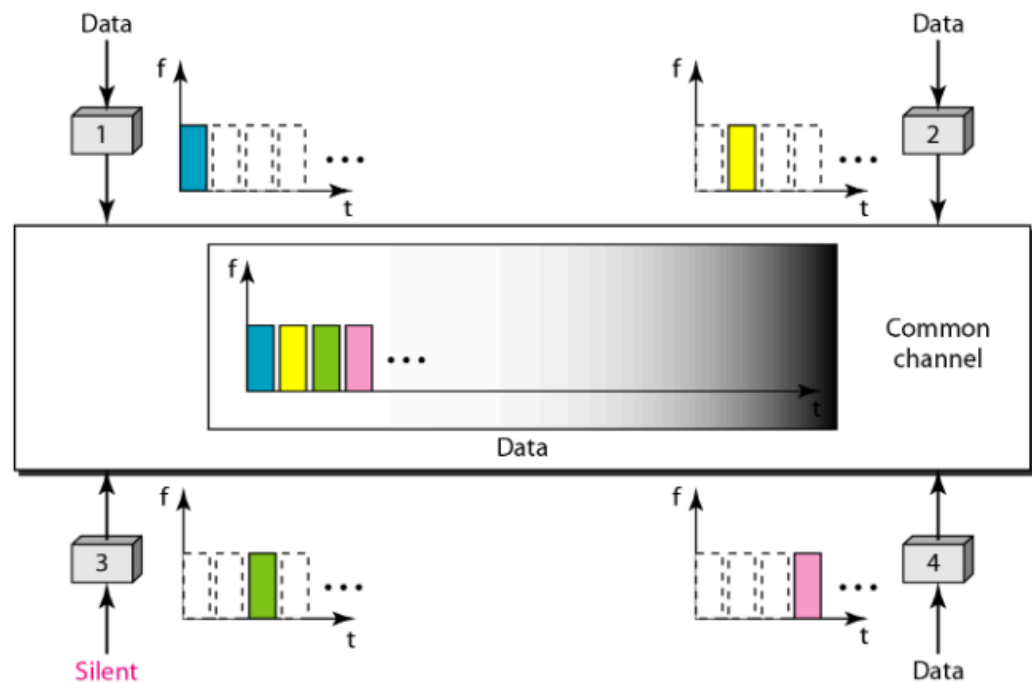
Time-Division Multiple Access (TDMA)

Code-Division Multiple Access (CDMA)

3.1 TDMA

In TDMA, the bandwidth is just one channel that is timeshared between different stations.

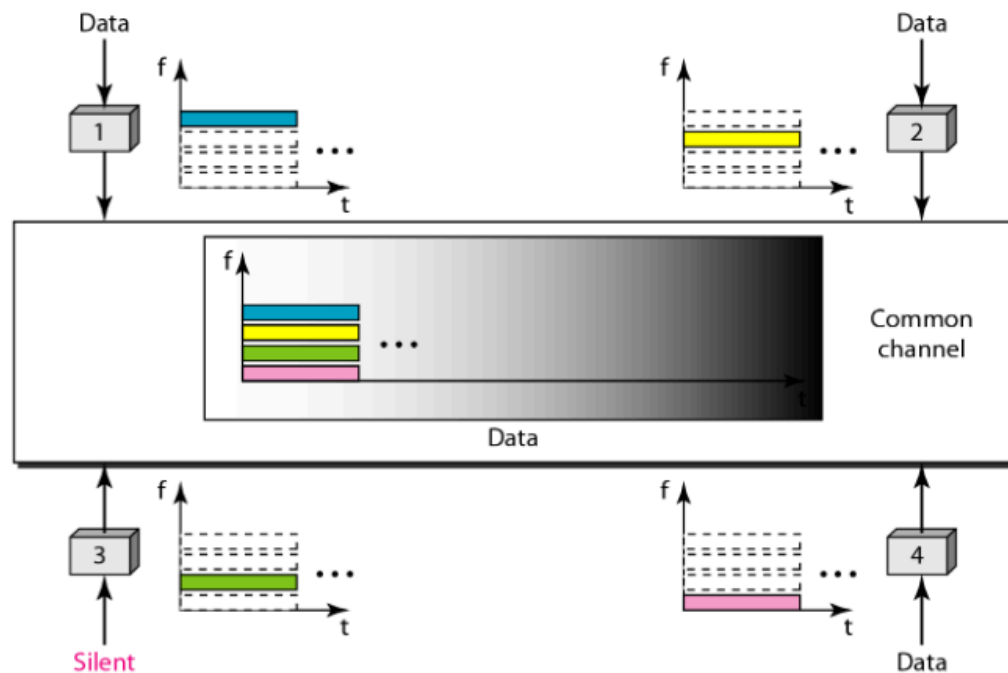
Figure 12.22 *Time-division multiple access (TDMA)*



3.2 FDMA

In FDMA, the available bandwidth of the common channel is divided into bands that are separated by guard bands.

Figure 12.21 *Frequency-division multiple access*
(FDMA)



3.3 SDMA

3.4 CDMA

In CDMA, one channel carries all transmissions simultaneously.

Figure 12.23 *Simple idea of communication with code*

